# Methodology Updates for the Vintage 2024 Estimates

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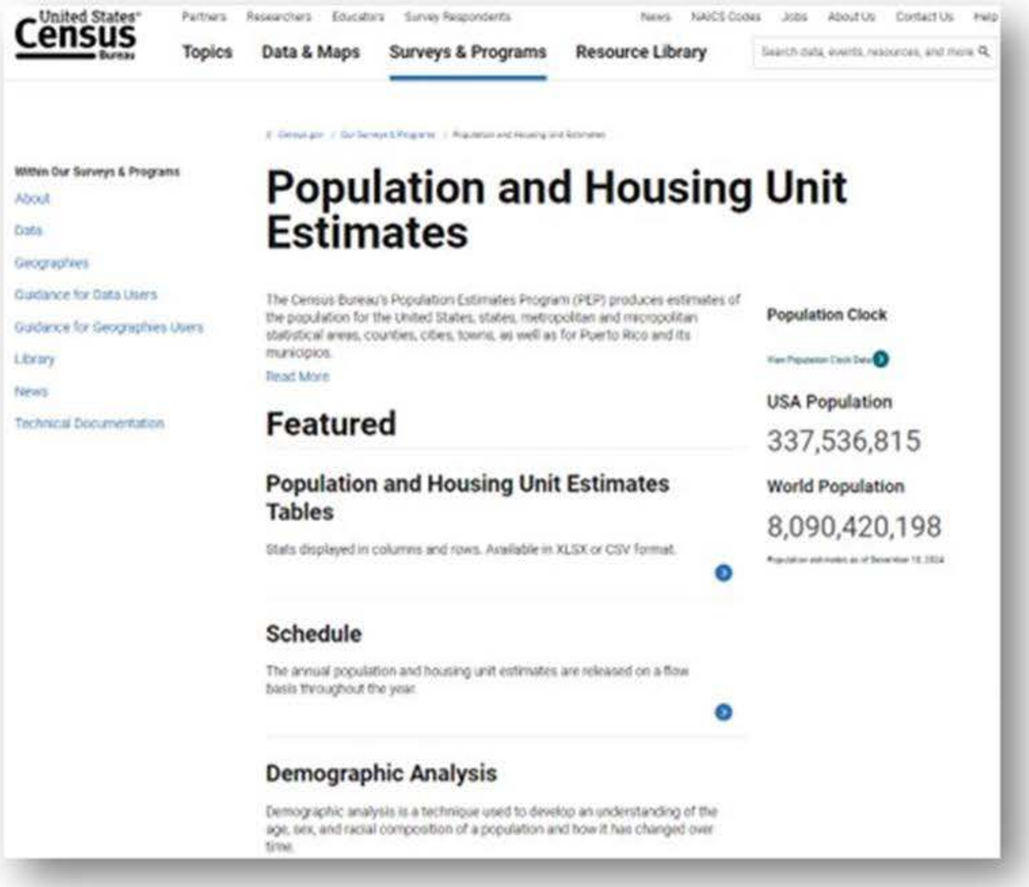
December 16, 2024

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*This presentation is released to inform interested parties of ongoing research and to encourage discussion of work in progress.*
*Any opinions and conclusions expressed herein are those of the author(s) and do not reflect the views of the U.S. Census Bureau.*

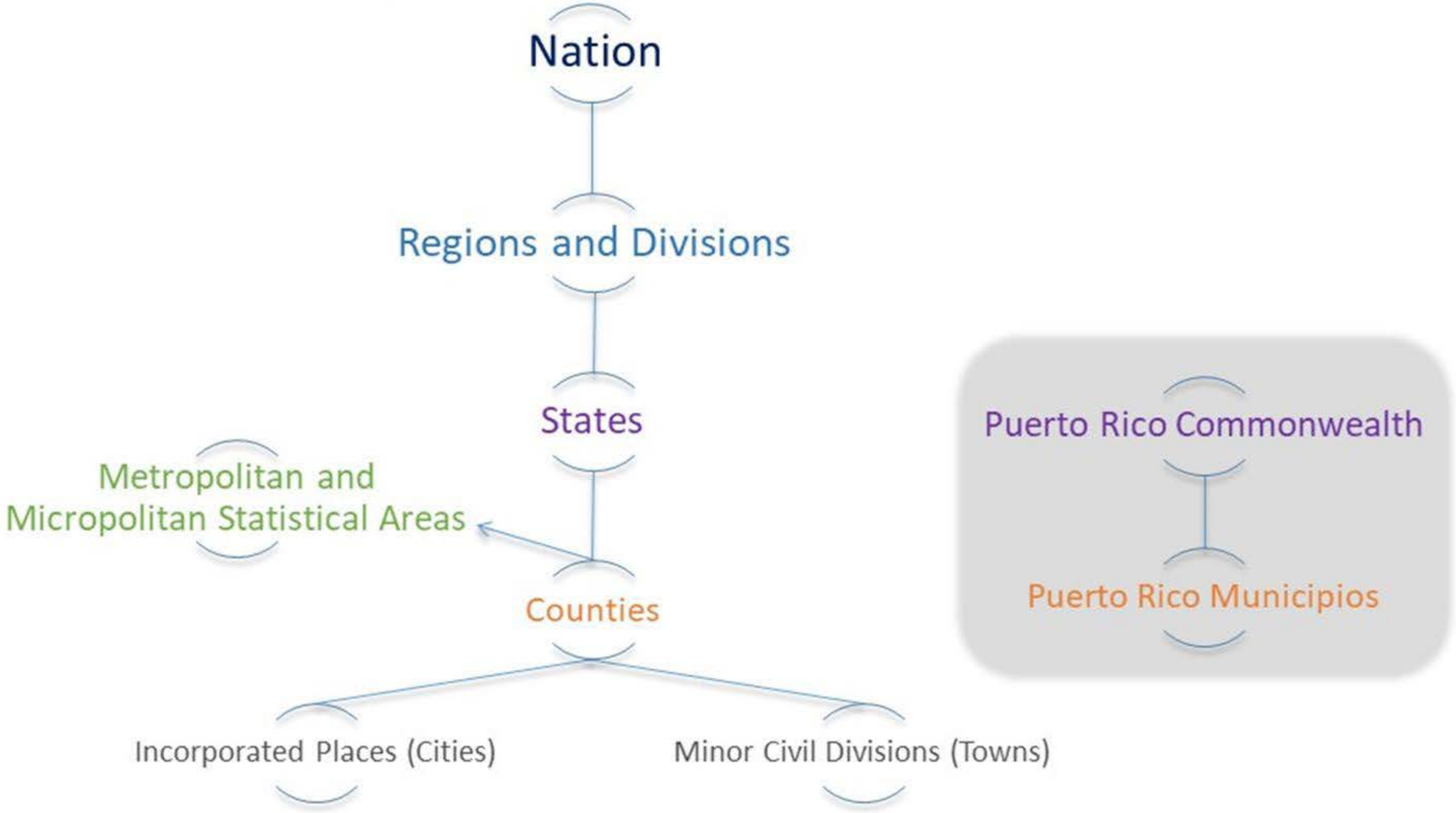
# The Population Estimates Program

- The Population Estimates Program (PEP) disseminates official measures of population and housing units between decennial censuses
- Developed as per Title 13, U.S. Code, Section 181
- Use cases include:
  - Population controls and denominators
  - Academic and business research
  - Program planning in the public and private sectors
- Time series are released annually featuring data for the date of the last census through the vintage year, which represents the latest year of estimates available
- Vintage 2023 is the current series
  - Contains estimates for April 1, 2020 to July 1, 2023




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# PEP Estimates Geographies




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**80,000+ Geographic Areas / 40,000+ Communities**

# Annual Estimates Detail

| Estimate Type | U.S. | States | Metro/ Micro Statistical Areas | Counties | Cities and Towns | Puerto Rico | Puerto Rico Municipios |
|---|---|---|---|---|---|---|---|
| Total Resident Population | X | X | X | X | X | X | X |
| Components of Change (Births, Deaths, Net Migration) | X | X | X | X | | X | |
| Population Characteristics (Age, Sex, Race, Hispanic Origin*) | X | X | X | X | | X | X |
| Monthly Population (Five Universes**) | X | | | | | | |
| Group Quarters Population | X (by char.) | X (totals) | | X (totals) | | | |
| Housing Units | X | X | | X | | | |

*For Puerto Rico Commonwealth and its municipios, population estimates are produced by age and sex only.
**The five estimates universes are: resident, resident plus Armed Forces overseas, civilian, civilian noninstitutionalized, household.

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# Availability of Population and Housing Unit Data Products: Vintage 2024

| Release Date* | Product | Geography |
|---|---|---|
| December 2024 | Population totals, components of change, and voting-age population | Nation, states, and Puerto Rico Commonwealth |
| March 2025 | Population totals and components of change | Metropolitan and micropolitan statistical areas, counties, and Puerto Rico municipios |
| April 2025 | Population by age and sex | Nation |
| May 2025 | Population totals-- | Cities and towns (incorporated places, minor civil divisions) |
| | Housing unit totals-- | Nation, states, and counties |
| June 2025 | Population by age, sex, race, and Hispanic origin-- | Nation, states, metropolitan and micropolitan statistical areas, and counties |
| | Population by age and sex-- | Puerto Rico Commonwealth and municipios |

*Dates subject to change.

census.gov/programs-surveys/popest/about/schedule.html

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# How to Access Population Estimates Datasets

Census.gov homepage

Surveys & Programs

Population Estimates (census.gov/popest)

Data

**Population and Housing Unit Estimates Datasets**

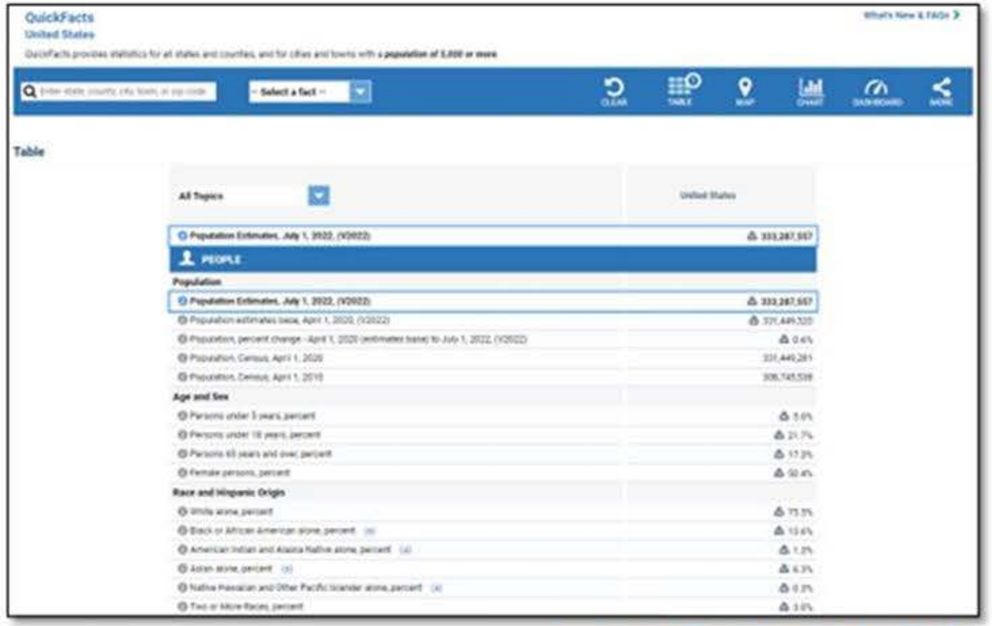
Datasets display population estimates totals and demographic characteristics. Datasets are in downloadable CSV format.

**Population and Housing Unit Estimates Tables**

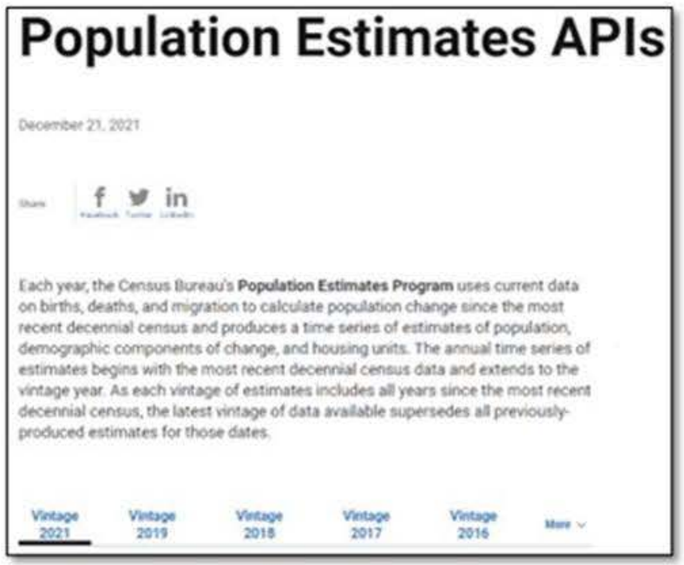
Stats displayed in columns and rows. Available in XLSX or CSV format.

**Population and Housing Unit Estimates Data Tools**

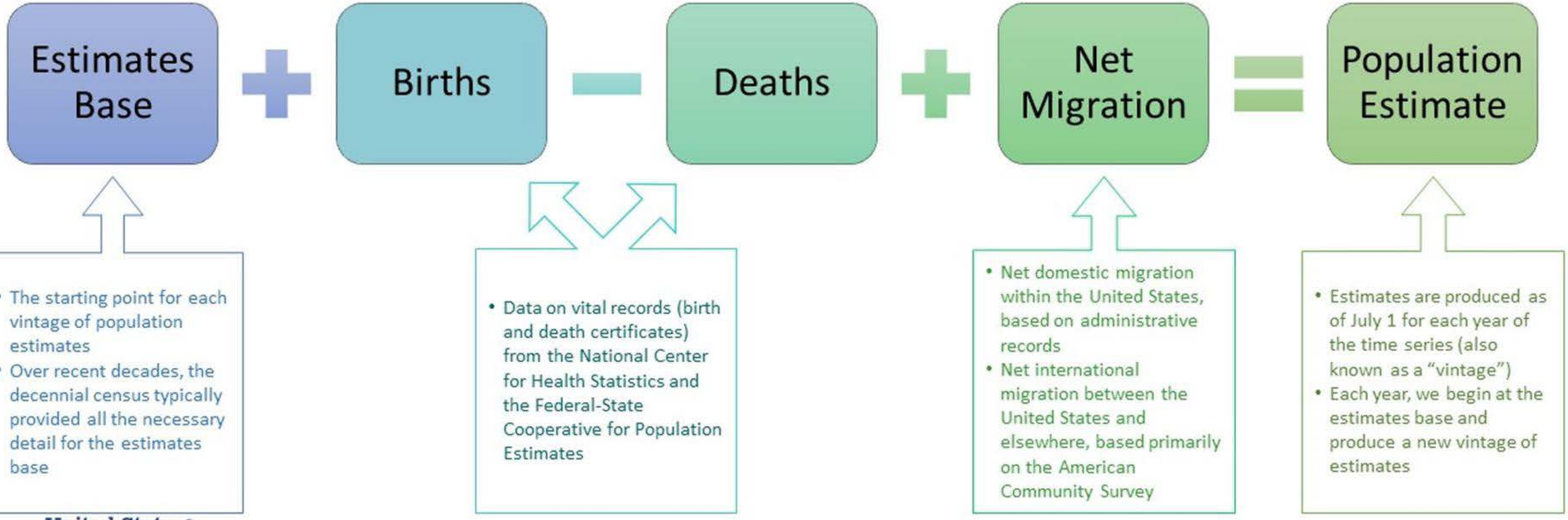
Interactive applications, created by the Census Bureau, to help you customize and visualize statistics from multiple censuses, surveys, and programs.




census.gov/quickfacts/



# National, State, and County Estimates: *Cohort-Component Method*




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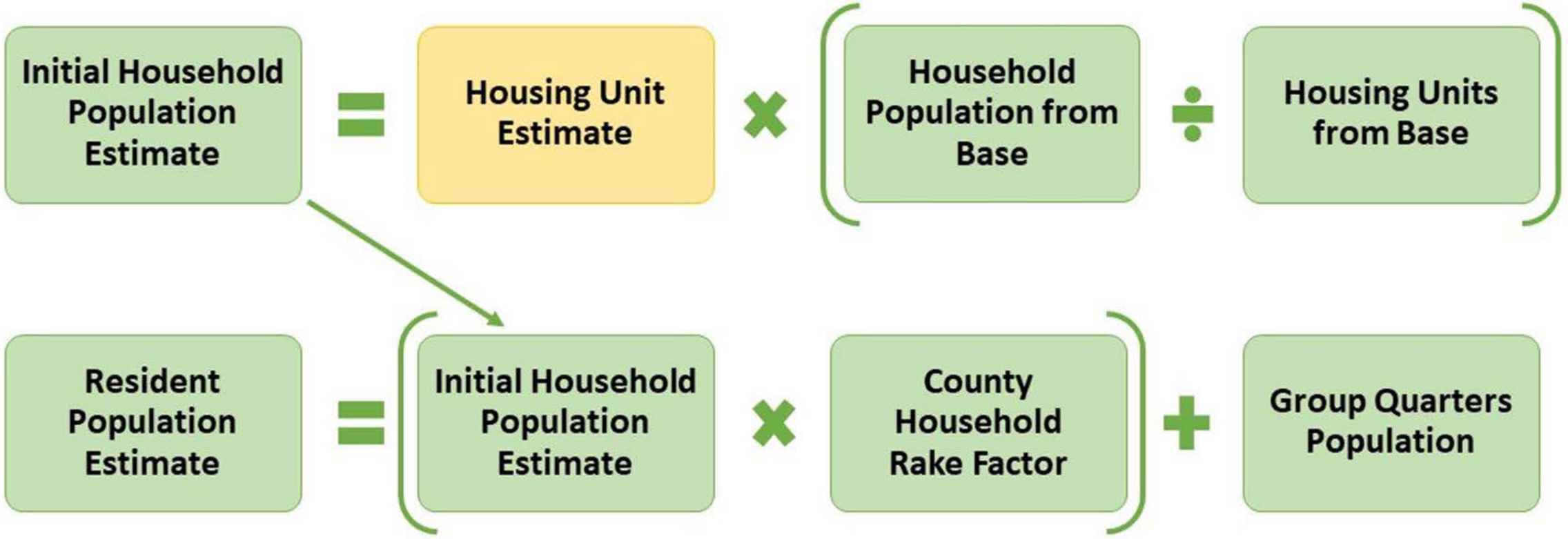
# Housing Unit Estimates:
## *Components of Change Method*




- Components of change method measures changes to the housing stock since the last census using a variety of survey and administrative data
- Housing unit base is created from invariant 2020 Census housing unit counts

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# Subcounty Population Estimates:
## *Distributive Housing Unit Method*



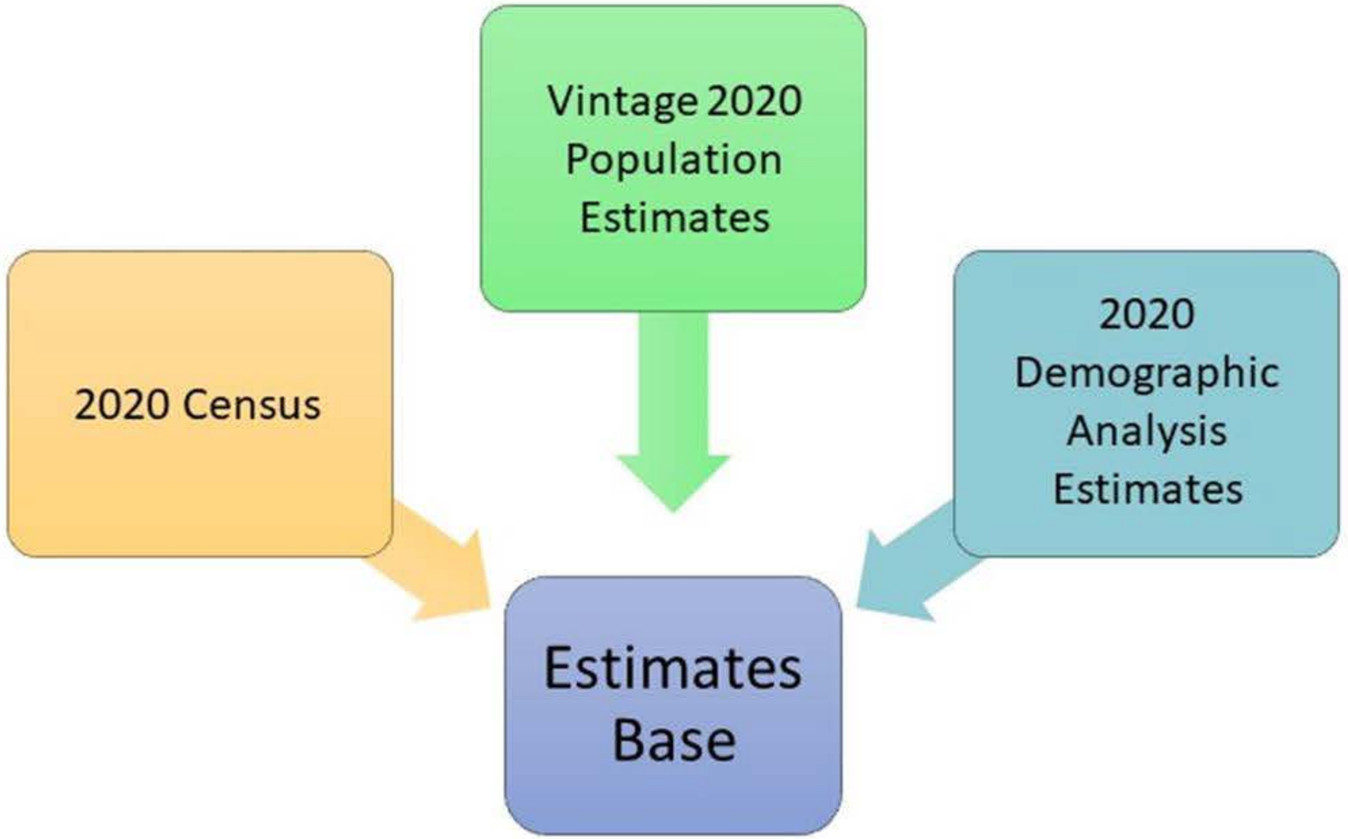

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# Annual Updates to the Estimates

- Base population updates: Legal boundary changes and other geographic updates, outcomes from official census programs (e.g., Count Question Resolution), changes to the methods of estimating the base
- Revisions to the components of change: Incorporation of more recent and/or complete data, changes to the methods of estimating the components of population change and special population stock estimates used in the estimates process
- Changes to the method of estimating population: Revisions to the method of estimating population using the components of change and special population stock estimates

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# Developing the Post-2020 Estimates Base




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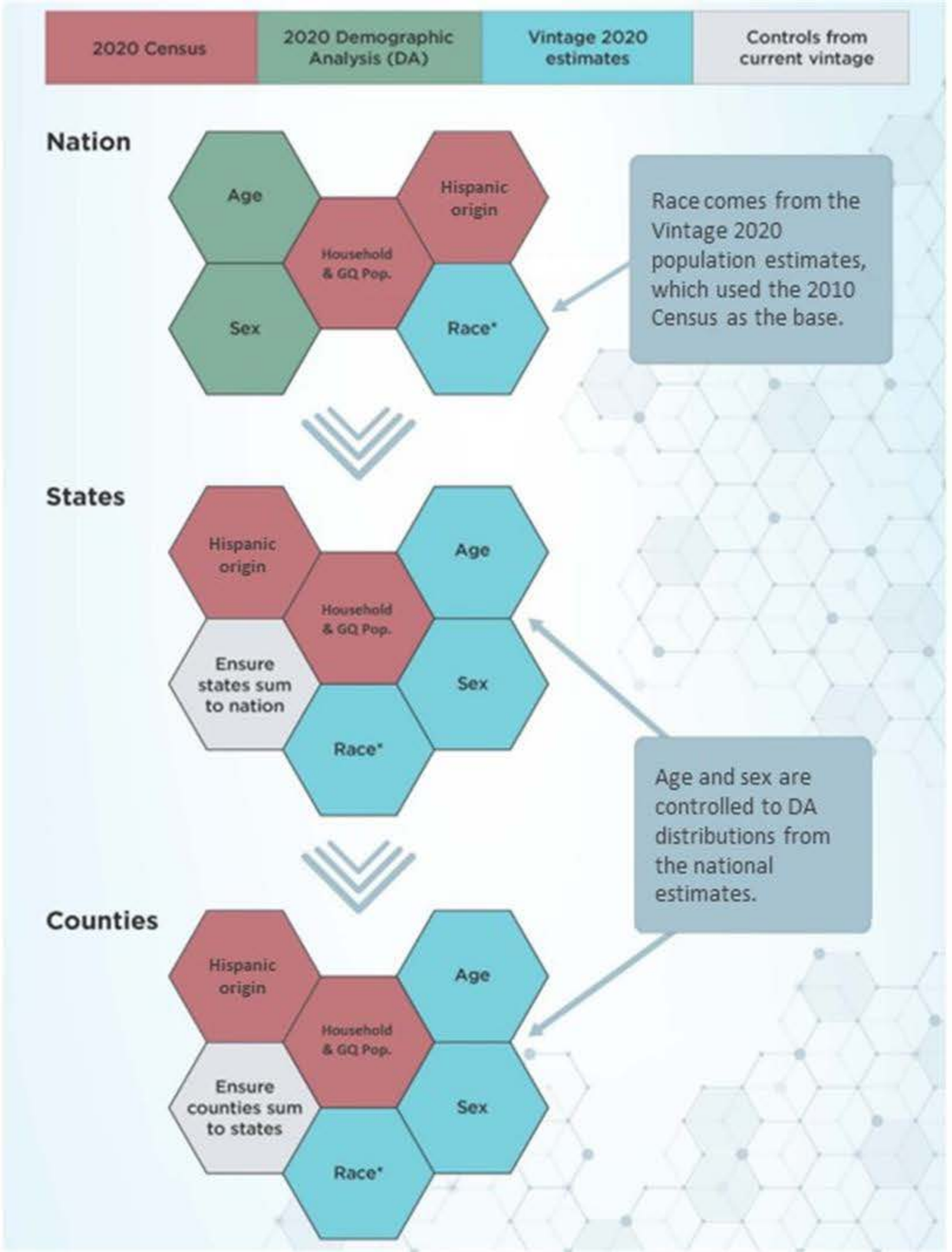
# Impacts of the Blended Base

- Original iteration (Vintage 2021):
  - Resulted in a smoother age/sex distribution than the 2020 Census counts.
  - Had a mitigating effect on the persistent undercount of young children in the census, particularly 0 to 4-year-olds (undercounted in the 2020 Census) and 10 to 14-year-olds (undercounted in the 2010 Census and aged forward in the estimates methodology)
- Subsequent iterations:
  - Captured geographic updates
  - Improved the processing of group quarters population data
  - Incorporated resident population by Hispanic origin from the 2020 Census

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# Blended Base Approach for Vintage 2024

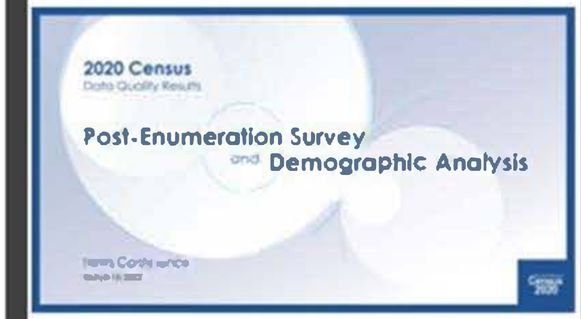
- **2020 Census Data from the Internal Census Edited File**:
  - Tabulated into 2024 geographies at the subcounty level, infused with differentially private noise, and then aggregated to create county, state, and national total resident, household, and group quarters (by facility type) population counts
  - County, state, and national Hispanic or Latino origin from the 2020 Census PL 94-171 Redistricting File for the total resident population
- **2020 Demographic Analysis (DA) Estimates**: National population estimates by age and sex
- **Vintage 2020 Population Estimates for April 1, 2020**:
  - Nation, state, and county population estimates by age, sex, and race
  - Hispanic origin, used to distribute the household and group quarters (by facility type) populations

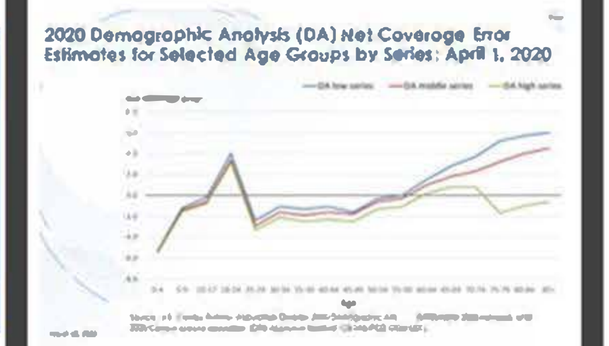



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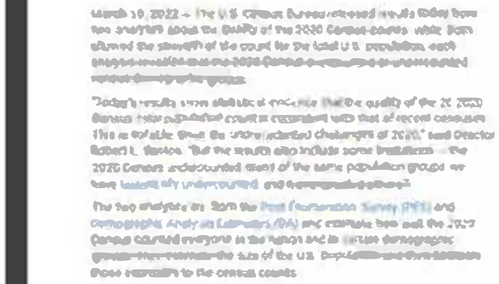
For Immediate Release: Thursday, March 10, 2022

# Census Bureau Releases Estimates of Undercount and Overcount in the 2020 Census




## Post-Enumeration Survey and Demographic Analysis Help Evaluate 2020 Census Results



## Coverage of the Census

| | |
|---|---|
| Census count of the population | 323,200,000 |
| Estimate of the population from the PES | 323,900,000 |
| Net coverage error | -782,000 |
| Percent net coverage error | -0.24 % |




## Demographic Analysis Method

Population (=) Births (-) Deaths (+) Immigration (-) Emigration

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# 2020 Census Coverage Concerns

- Following the release of 2020 Census coverage measures, concerns about census quality became apparent
- Many looked to the population estimates, and particularly the blended base, as a potential solution
  - Stakeholders, advisory groups, census leadership, media

# The Base Evaluation and Research Team (BERT)

- BERT has been meeting since March 2022, focusing research on:

| **2020 Census data by subject area (age/sex, Hispanic origin, race)** | **2020 Census Coverage Measures** |
|---|---|
| *Analyses*<br>• Comparisons to benchmark data<br>• Evaluations of change since the 2010 Census<br>• Assessment of demographic plausibility<br>*Key research questions*<br>• Do these data improve the blended base?<br>• Are there adjustments that can be applied to these data to address coverage issues (before or after they are incorporated into the base)? | *Analysis*<br>• Assessment of methodology<br>*Key research questions*<br>• Based on their design, are these measures fit for use in the estimates base?<br>• Are there design changes (i.e., recommendations for 2030) which would increase the utility of these measures for the estimates base? |

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# Summary of Updates to the Methodology for Vintage 2024

- Base Population
  - **United States** – no changes relative to Vintage 2023
  - **Puerto Rico** – now uses group quarters population characteristics (counts by facility type from the 2020 Census PL 94-171 Redistricting File)
- Vital Statistics
  - We began a phased transition to estimating births and deaths based exclusively on data from the National Center for Health Statistics (NCHS) at the subnational level
  - Previously, we combined data provided by representatives of the Federal-State Cooperative for Population Estimates (FSCPE) with birth and death records from NCHS
    - FSCPE data were used to adjust the geographic distribution of births and deaths within the state

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# Vintage 2024 Updates to Net International Migration

- In the last three years, there has been a notable increase in the number of refugees, U.S. Border Patrol releases, and U.S. Customs and Border Protection Office of Field Operations' parolees
- We used newly available and reliable monthly administrative data on these humanitarian migrant cases from the U.S. Refugee Admissions Program and the Office of Homeland Security Statistics to enhance the estimates of non-U.S.-born immigration
- An adjustment was applied at the national level by adding 75% of the humanitarian migrants in our benchmark database to the American Community Survey (ACS) residence one year ago non-U.S.-born immigration estimate for each estimate year
- This is similar to the adjustments for the impact of COVID-19 that were implemented for the V2021 and V2022 estimates
- The adjustment for humanitarian migrants is incorporated into the national total only
    - Forthcoming research will focus on the distribution of the adjustment for subnational geographies and by demographic characteristics

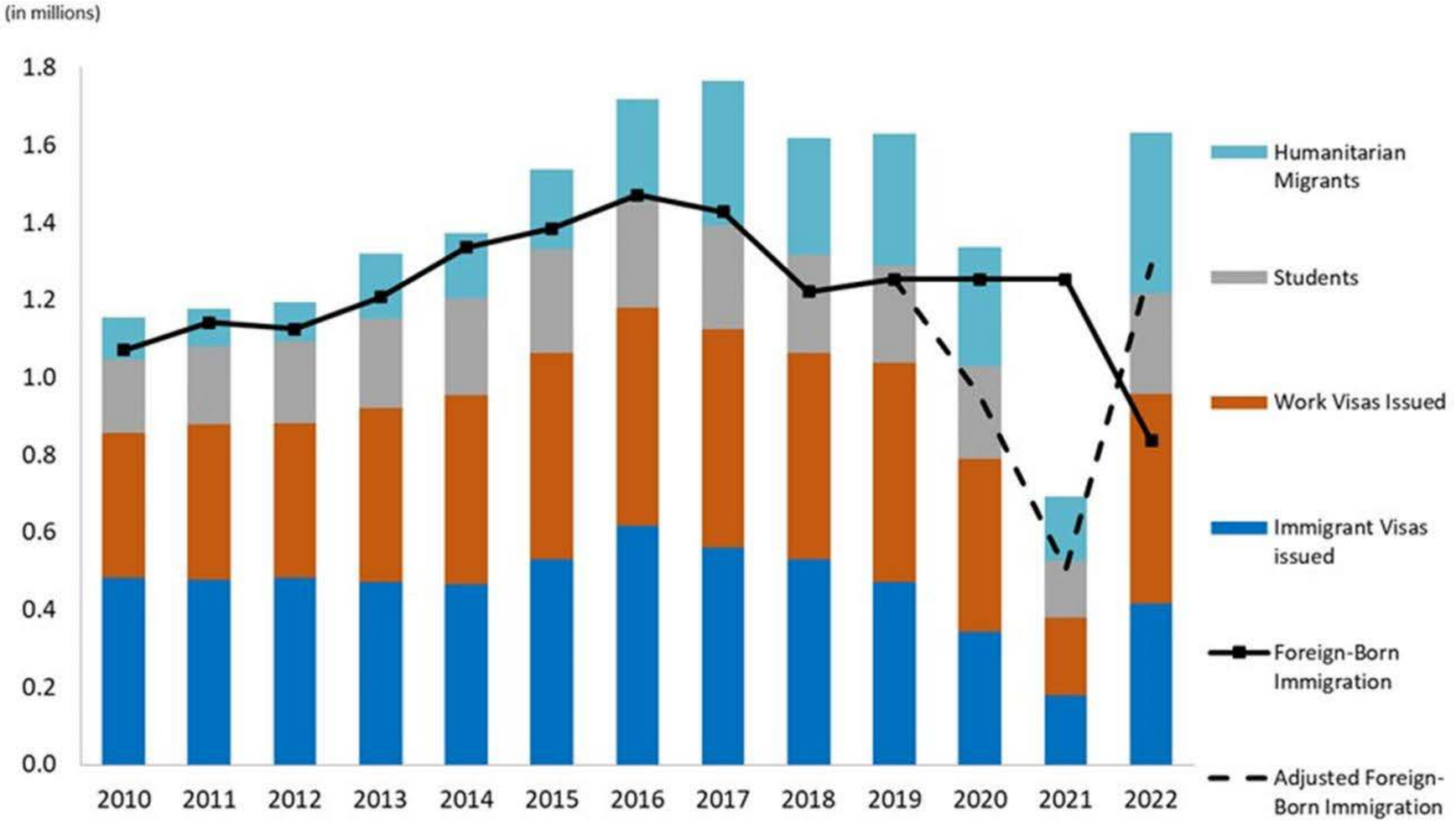
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# Vintage 2024 Updates to Net International Migration (cont.)

- This adjustment addresses two methodological limitations resulting from how we use the ACS to estimate international migration flows
  1. The ACS data that we use are lagged by one year when we are producing the population estimates
     - For example, the 2023 1-Year ACS was the most current file available when we produced the V2024 population estimates
  2. The foreign-born population may be underrepresented in the ACS data, which would translate into underestimation of non-U.S.-born immigration flows
     - Several studies have found substantial coverage issues for some foreign-born groups
     - Temporary migrant shelters were not part of the ACS sampling frame

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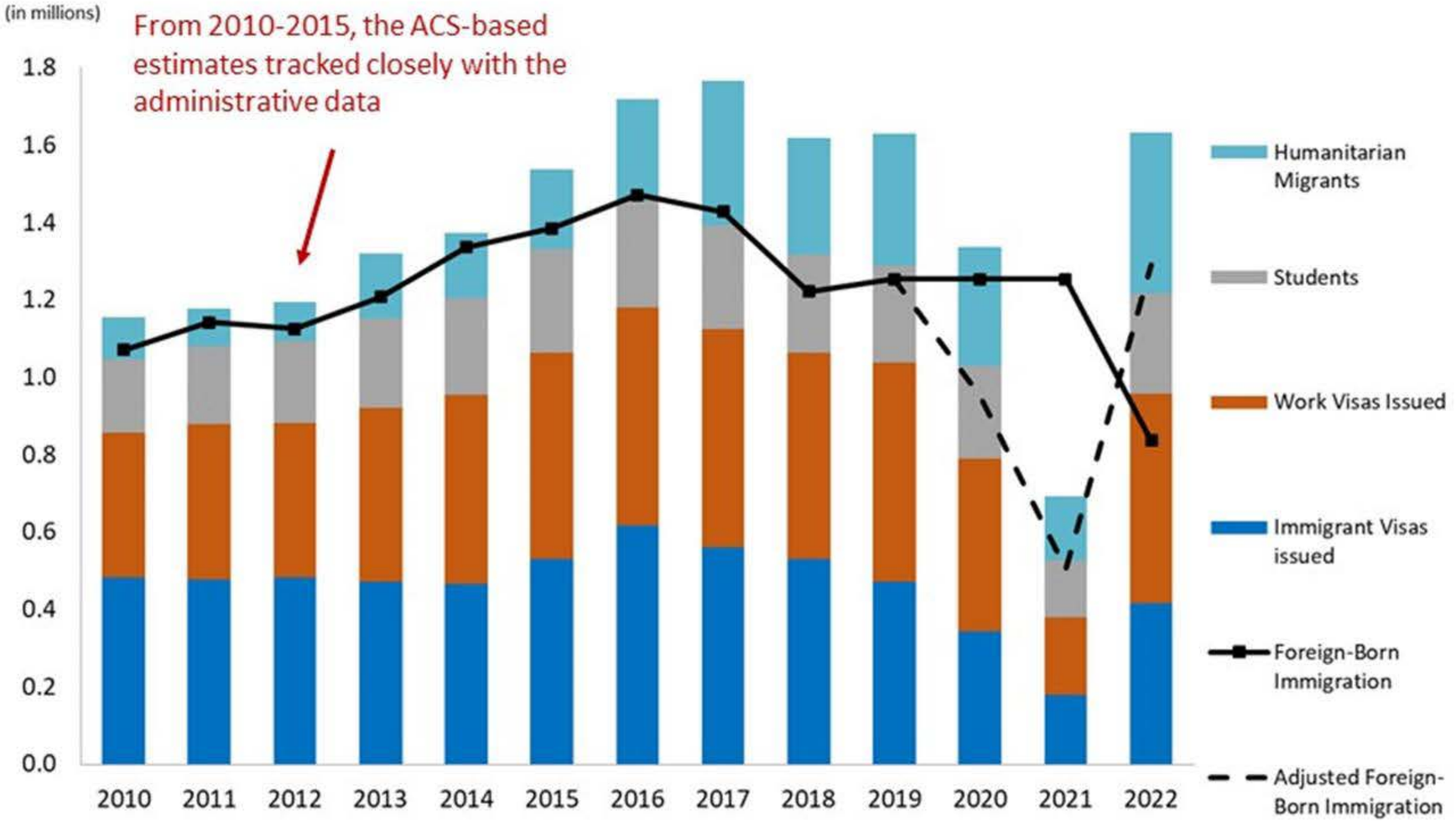
## Foreign-Born Immigration estimates and Select Administrative Data: 2010 to 2022




Source: U.S. Census Bureau Population Estimates Internal Research; U.S. Department of Justice; Institute of International Education; U.S. Citizenship and Immigration Services; U.S. Department of Homeland Security; and U.S. State Department Bureau of Consular Affairs and Refugee Processing Center.

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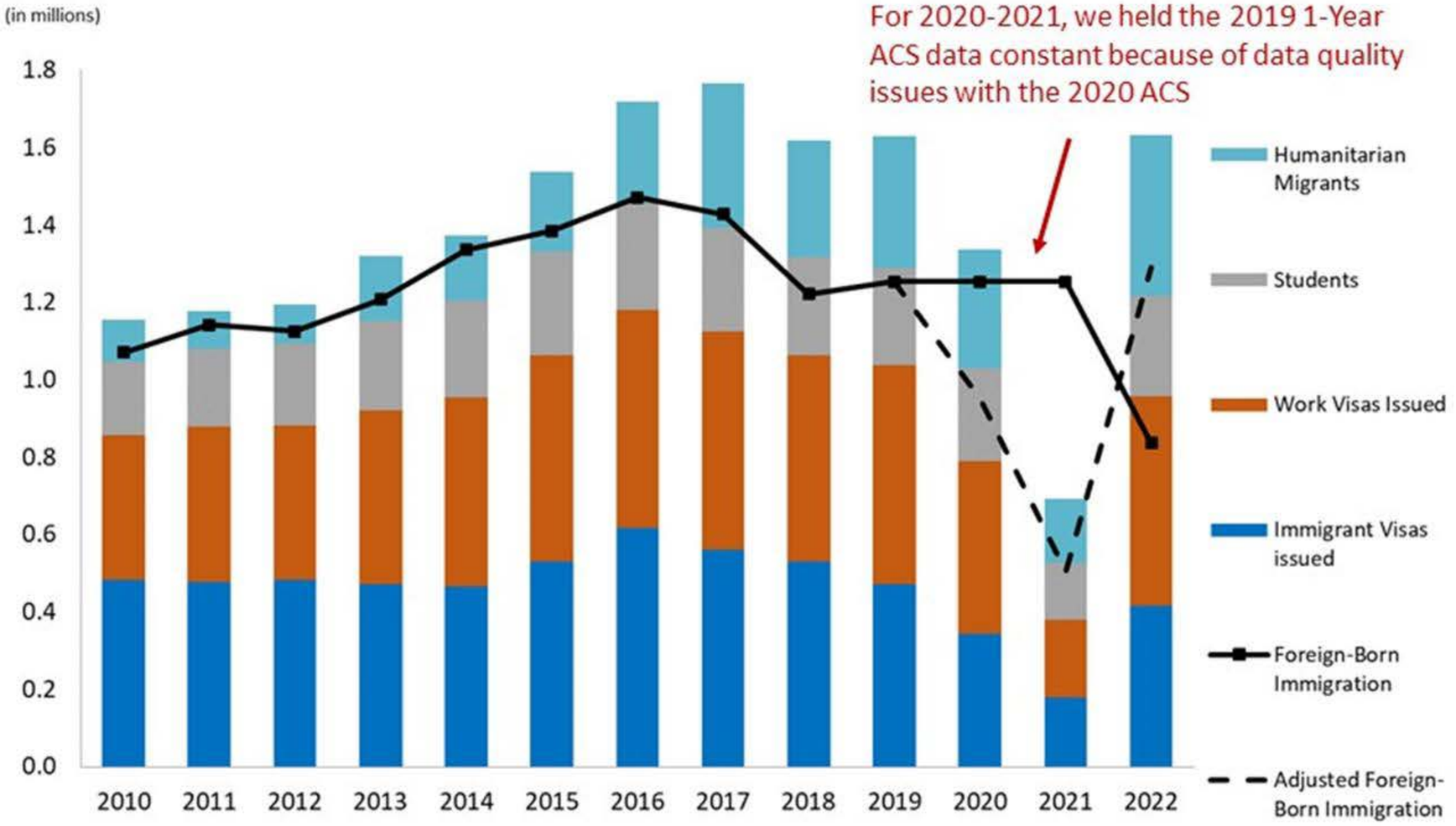
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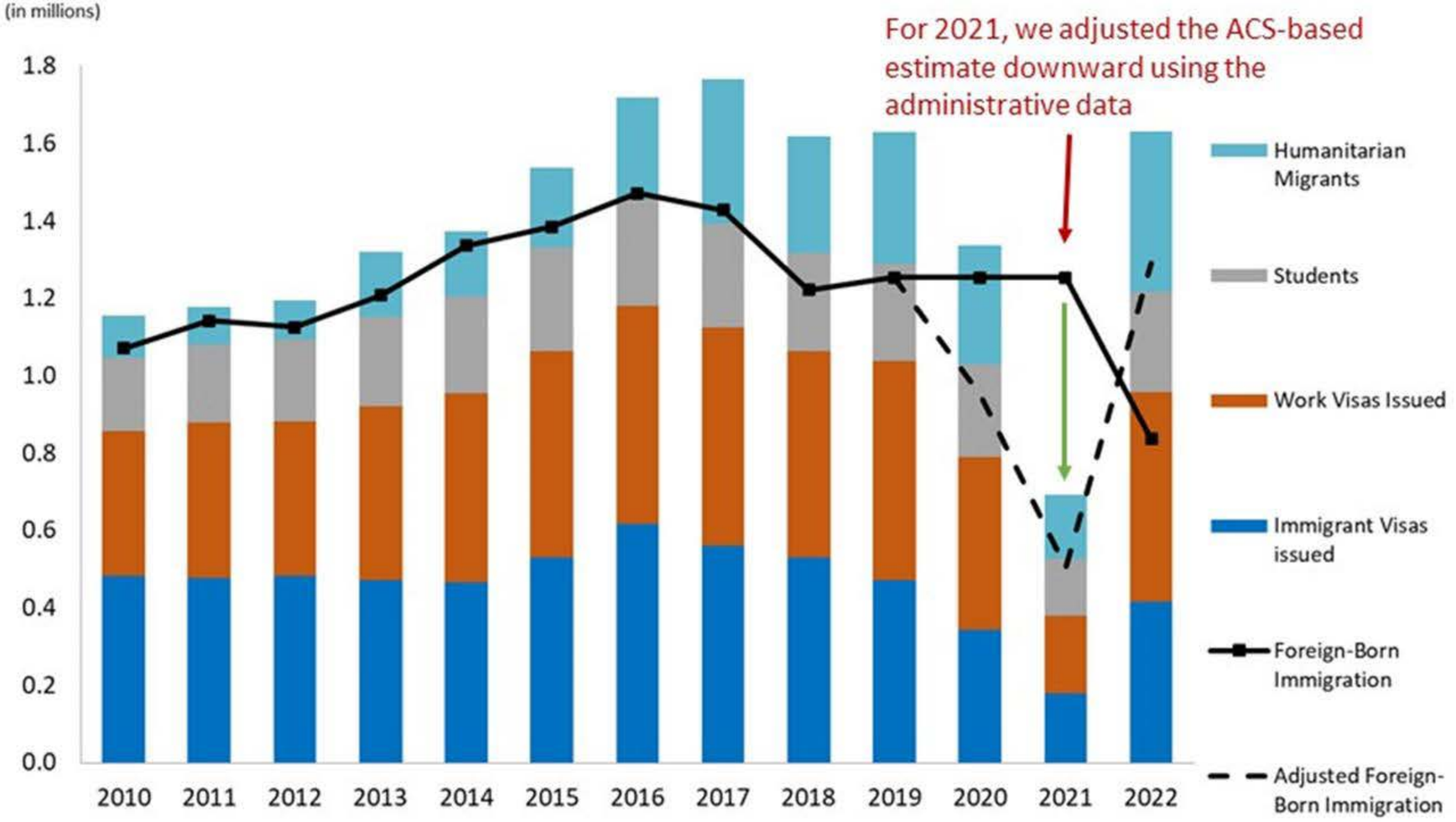
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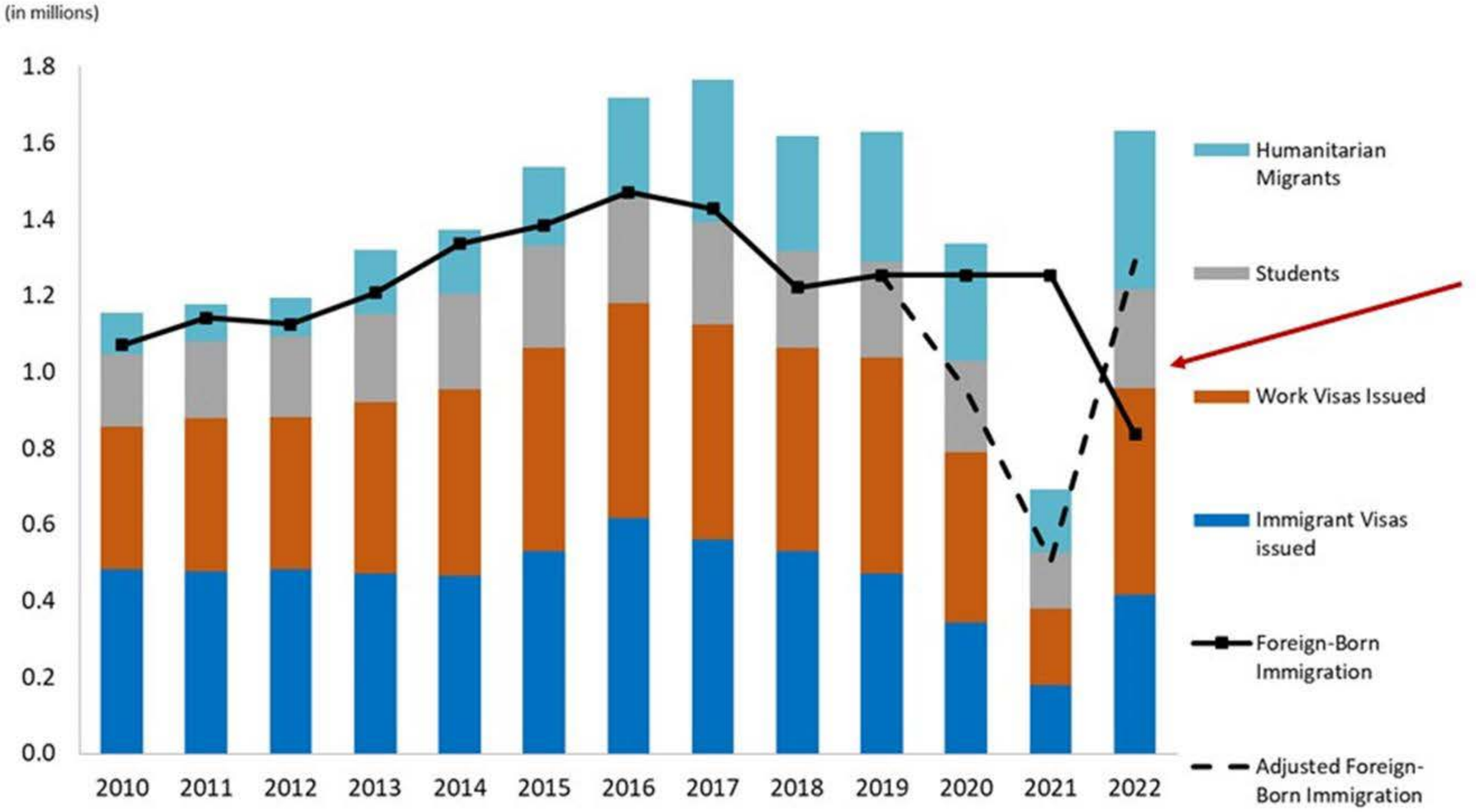
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## Foreign-Born Immigration estimates and Select Administrative Data: 2010 to 2022

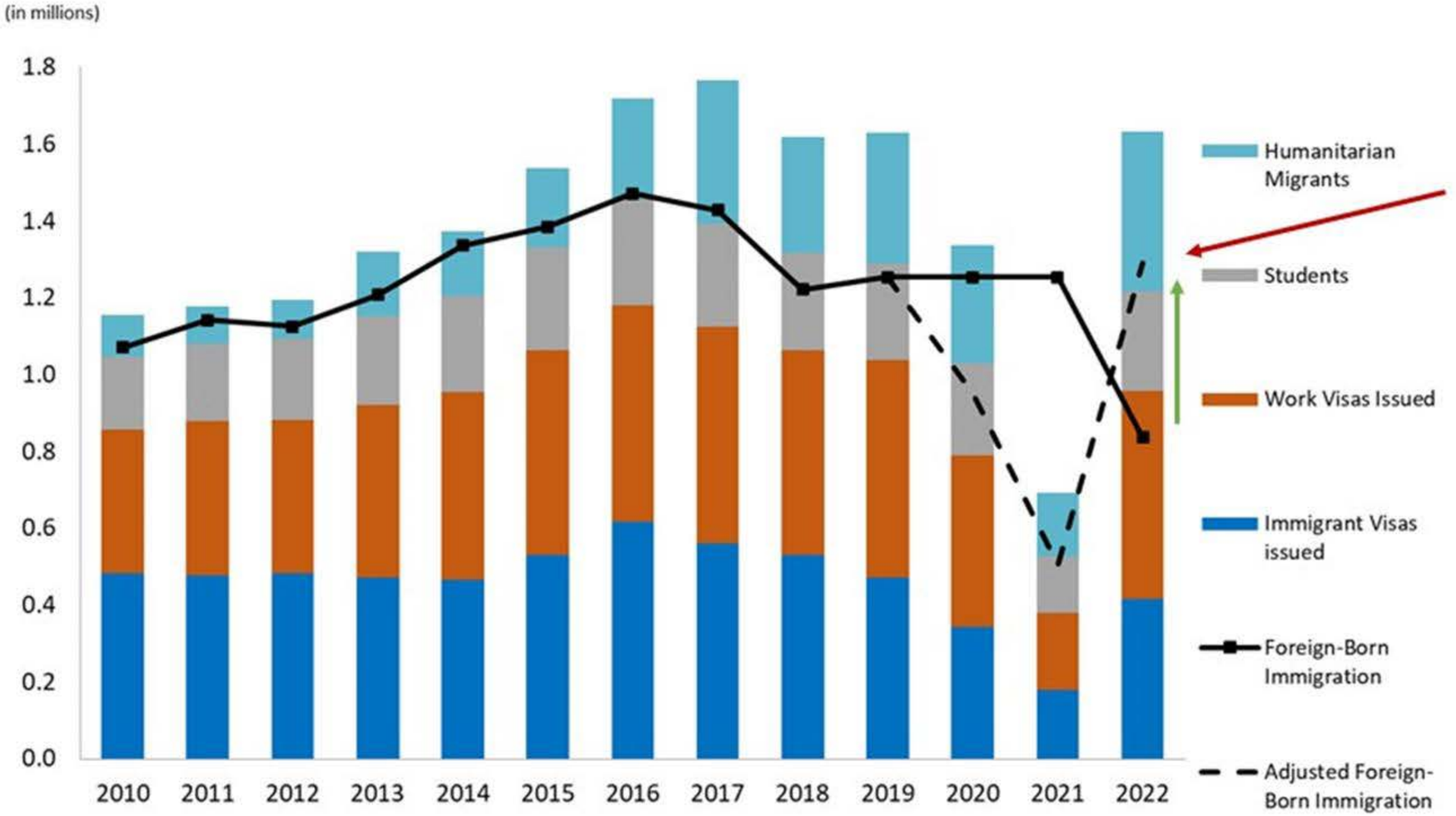



For 2022, the 2021 ACS 1-year file was the most recent survey data available

Source: U.S. Census Bureau Population Estimates Internal Research; U.S. Department of Justice; Institute of International Education; U.S. Citizenship and Immigration Services; U.S. Department of Homeland Security; and U.S. State Department Bureau of Consular Affairs and Refugee Processing Center.

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## Foreign-Born Immigration estimates and Select Administrative Data: 2010 to 2022




For 2022, we used the administrative data to adjust the estimate upward

Source: U.S. Census Bureau Population Estimates Internal Research; U.S. Department of Justice; Institute of International Education; U.S. Citizenship and Immigration Services; U.S. Department of Homeland Security; and U.S. State Department Bureau of Consular Affairs and Refugee Processing Center.

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# 2020 Modified Age & Race Census File

- We are planning to release the 2020 Census Modified Age & Race Census (MARC) file in early 2025
  - Some Other Race (SOR) observations are reallocated into the official Office of Management and Budget race categories
  - We need this file to use 2020 Census race data in the population estimates
  - The file also includes smoothed age data for a subset of observations to address “age heaping” in the 2020 Census
- **For Vintage 2025, we will use the 2020 MARC file to expand the amount of 2020 Census data in the blended base**

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# For More Information

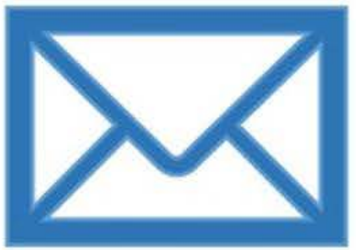
Coordination, Dissemination, and Outreach Branch
pop.cdob@census.gov, (301) 763-2110

Base Evaluation and Research Team
pop.bert@census.gov

Population and Housing Unit Estimates Website
https://www.census.gov/programs-surveys/popest.html

Blog on Age Heaping in the 2020 Census
https://www.census.gov/newsroom/blogs/random-samplings/2023/05/age-heaping-2020-census-dhc.html

Blog on BERT Recommendations Regarding the Use of the 2020 PES in the Vintage 2023 Population Estimates
https://www.census.gov/newsroom/blogs/random-samplings/2023/12/recommendations-2020-pes-coverage-results-in-vintage-2023-pop-estimates.html



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# Questions



**Thank You!**

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